

HDOC Day-19

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

Aim

To write a C program that calculates the Semester Grade Point Average (SGPA) for each semester and the final Cumulative Grade Point Average (CGPA), with validation of invalid and boundary inputs.

Theory / How the code is done

The program accepts a positive Global ID and the number of semesters. The outer **for loop** handles semesters and the inner **for loop** handles courses. **do-while loops** validate the ID, number of semesters, number of courses, credits and grades. A **switch-case** converts each grade to a grade point. For each course, grade credits are $GP \times credit$. SGPA is semester total grade credits divided by semester total credits, while CGPA is cumulative grade credits divided by cumulative credits.

Grade Point Conversion

Grade	O	A	B	C	D	E	P
Grade Point	10	9	8	7	6	5	4

Algorithm

1. Start and declare all required integer, floating-point and character variables.
2. Read Global ID; if it is not positive, display an error and read it again.
3. Read m, the number of semesters; validate that m is positive.
4. Set cumulative_TC = 0 and cumulative_TGC = 0.
5. For each semester, set semester_TC = 0 and semester_TGC = 0.
6. Read n, the number of courses, and validate that n is positive.
7. For each course, read course code, positive credit and a valid grade.
8. Use switch-case to convert the grade: O=10, A=9, B=8, C=7, D=6, E=5, P=4.
9. Add $GP \times credit$ to semester_TGC and credit to semester_TC.
10. Calculate $SGPA = \text{semester_TGC} / \text{semester_TC}$ and display it.
11. Add semester totals to cumulative totals.
12. After all semesters, calculate $CGPA = \text{cumulative_TGC} / \text{cumulative_TC}$; display final details and stop.

C Program

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    long long global_id;
    int m, n, course, credit, GP;
    char code[20], grade;
    int semester_TC, cumulative_TC = 0;
    double semester_TGC, cumulative_TGC = 0.0;
    double SGPA, CGPA;

    do {
        printf("Enter Global ID: ");
        scanf("%lld", &global_id);
        if (global_id <= 0)
            printf("Invalid Global ID! Enter a positive value.\n");
    } while (global_id <= 0);

    do {
        printf("Enter number of semesters: ");
        scanf("%d", &m);
        if (m <= 0)
            printf("Invalid number of semesters! Enter a positive value.\n");
    } while (m <= 0);

    for (int i = 1; i <= m; i++)
    {
        semester_TC = 0;
        semester_TGC = 0.0;
        printf("\n--- Semester %d ---\n", i);

        do {
            printf("Enter number of courses: ");
            scanf("%d", &n);
            if (n <= 0)
                printf("Invalid number of courses! Enter a positive value.\n");
        } while (n <= 0);

        for (course = 1; course <= n; course++)
        {
            printf("\nCourse %d\n", course);
            printf("Enter course code: ");
            scanf("%19s", code);

            do {
                printf("Enter credit: ");
                scanf("%d", &credit);
                if (credit <= 0)
                    printf("Invalid credit! Enter a positive value.\n");
            } while (credit <= 0);

            do {
                printf("Enter grade (0/A/B/C/D/E/P): ");
                scanf(" %c", &grade);
                grade = toupper((unsigned char)grade);

                switch (grade) {
                    case 'O': GP = 10; break;
                    case 'A': GP = 9; break;
                    case 'B': GP = 8; break;
                    case 'C': GP = 7; break;
                    case 'D': GP = 6; break;
                    case 'E': GP = 5; break;
                    case 'P': GP = 4; break;
                    default:
                        GP = -1;
                        printf("Invalid grade! Enter O, A, B, C, D, E or P.\n");
                }
            } while (GP == -1);

            semester_TGC += GP * credit;
            semester_TC += credit;
        }

        SGPA = semester_TGC / semester_TC;
        printf("\nSemester %d Total Credits = %d\n", i, semester_TC);
        printf("Semester %d SGPA = %.2f\n", i, SGPA);

        cumulative_TGC += semester_TGC;
    }
}
```

```
        cumulative_TC += semester_TC;
    }

    CGPA = cumulative_TGC / cumulative_TC;

    printf("\n===== FINAL RESULT =====\n");
    printf("Global ID = %lld\n", global_id);
    printf("Number of Semesters = %d\n", m);
    printf("Cumulative Total Credits = %d\n", cumulative_TC);
    printf("CGPA = %.2f\n", CGPA);

    return 0;
}
```

Dry Run / Sample Calculation

Semester	Course	Credit	Grade	GP	Grade Credits
1	CSE101	4	O	10	40
1	MAT101	3	A	9	27
1	PHY101	3	B	8	24
2	CSE102	4	A	9	36
2	MAT102	4	B	8	32

Semester 1: TC=10, TGC=91, SGPA=9.10.

Semester 2: TC=8, TGC=68, SGPA=8.50.

Final: cumulative TC=18, cumulative TGC=159, CGPA=159/18=**8.83**.

Test Case Table

Case	Input / condition	Expected result
1	Global ID = -2, then 123456	Reject -2; accept positive ID
2	m = 0, then 2	Reject 0; accept 2
3	course count = -1, then 3	Reject -1; accept 3
4	credit = 0, then 4	Reject 0; accept 4
5	grade = X, then O	Reject X; accept O and GP=10
6	Valid sample data above	SGPA1=9.10, SGPA2=8.50, CGPA=8.83

Compilation and Execution

The program was compiled using **gcc sgpa_cgpa.c -o sgpa_cgpa**. Compilation completed successfully. The prepared sample was executed to verify the SGPA and CGPA calculations.

Sample Output

```
Enter Global ID: Enter number of semesters:
--- Semester 1 ---
Enter number of courses:
Course 1
Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P):
Course 2
Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P):
Course 3
Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P):
Semester 1 Total Credits = 10
Semester 1 SGPA = 9.10

--- Semester 2 ---
Enter number of courses:
Course 1
Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P):
Course 2
Enter course code: Enter credit: Enter grade (O/A/B/C/D/E/P):
Semester 2 Total Credits = 8
Semester 2 SGPA = 8.50

===== FINAL RESULT =====
Global ID = 123456
Number of Semesters = 2
Cumulative Total Credits = 18
CGPA = 8.83
```

Result

The program successfully calculates semester-wise SGPA and final CGPA using nested loops, do-while validation and switch-case decision making. The prepared valid and invalid test conditions are handled as required.